



Scrub-XR

*AI-powered surgical instrument tracking for
improving operating room efficiency and
patient outcomes*

Meet Crystal



Age: 51 year old

Residency: Long Island, NY

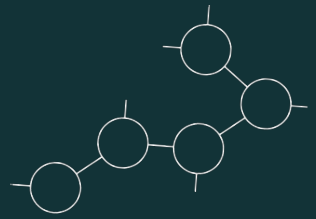
Procedure: Pacemaker replacement

Post-Op:

- Incision wasn't healing properly
- Chronic fever
- *"I could never make any type of plans even with my kids [...] I didn't know if I was going to be back in the hospital."*

What went wrong?

RSI (Retained Surgical Item)



***2" gauze left inside
her chest cavity***

- Led to **chronic infection**
- Required **2** corrective procedures
- **4** year recovery time



Surgical Counting



Problem Statement

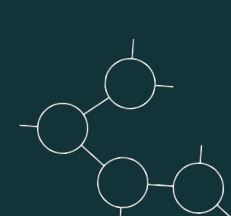
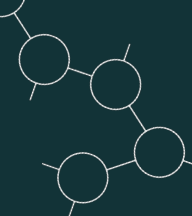
Surgical technicians are responsible for preparing the operating room and keeping track of surgical instruments during operations



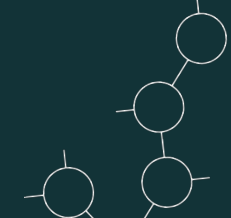
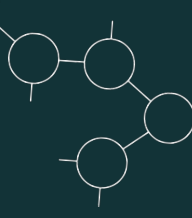
- **Surgical instrument counts are still performed manually!**
 - ◆ Occurs **4** times for each procedure
- ~ **2000** Retained Surgical Instrument (RSI) cases occur annually in the U.S.
 - ◆ RSI's are "never events"
 - ◆ **\$26 million** payout over 4 years
 - ◆ Re-Op costs are ~**\$78,000** for hospital

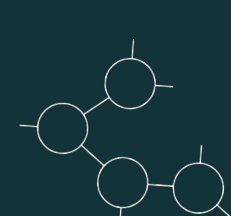
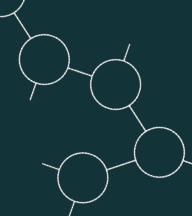
"The surgical counts are **time-consuming** and prone to **inconsistencies**"

- Caprice C. Greenberg, MD, MPH



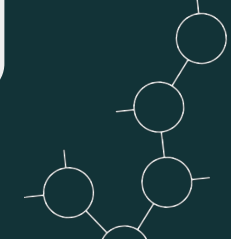
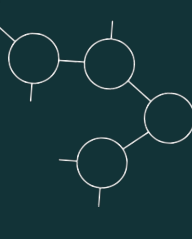
**How can we improve patient
safety and OR efficiency?**



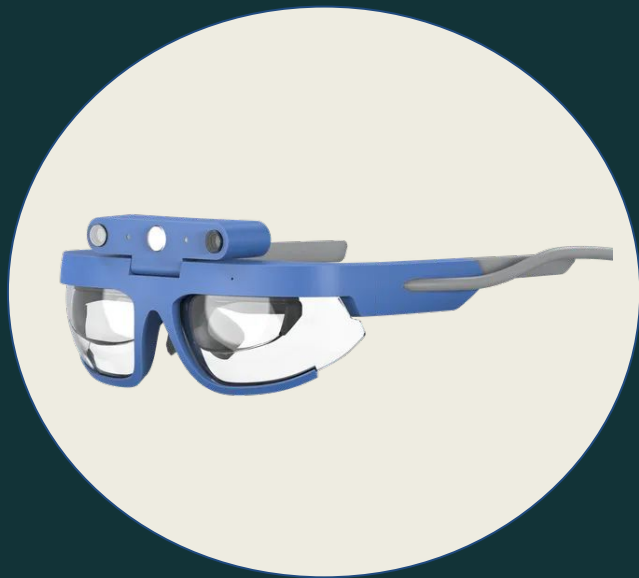


How can we improve patient safety and OR efficiency?

By utilizing XR to automate surgical instrument counts!



Our Solution



Automated Surgical Instrument Tracking

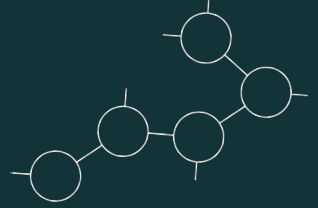
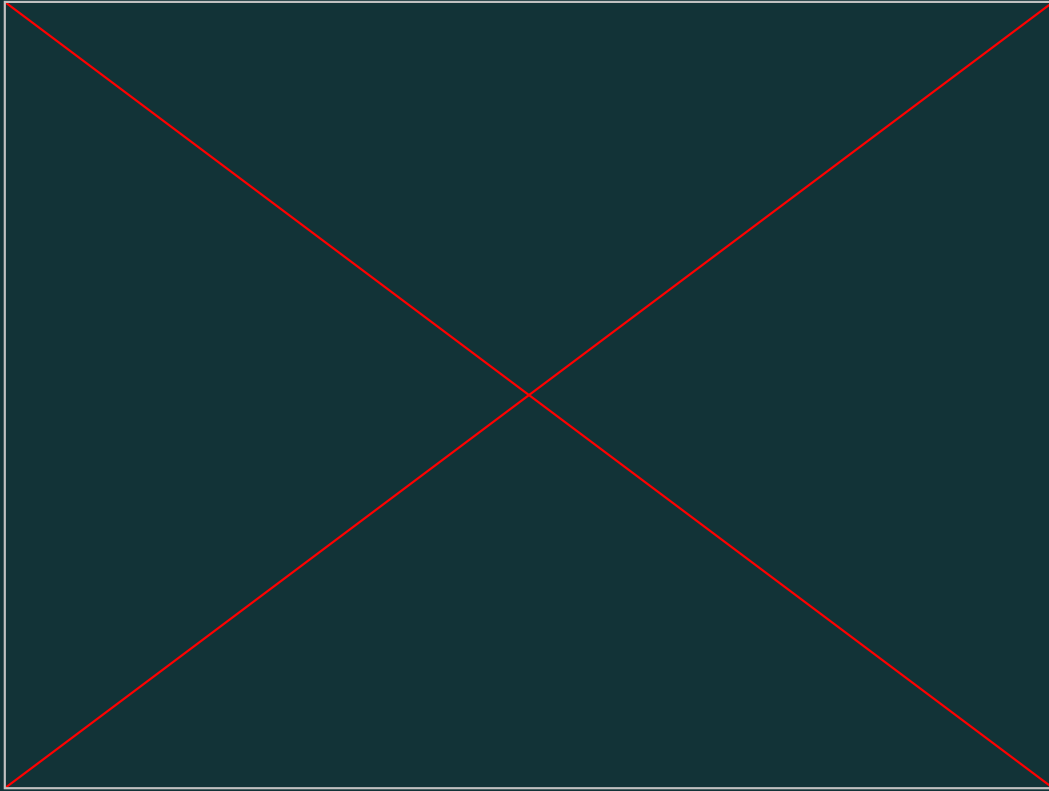
XR Glasses

- Computer vision identifies instruments and automatically updates counts in real time
 - ◆ Image-based counts at 4 key surgical stages
 - ◆ Real-time tracking as instruments are passed

Automatic dataset generated of instruments used per procedure



Demo



Results

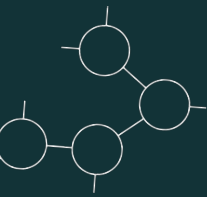
Training Instrument Set: scalpel, straight mayo scissors, curved mayo scissors, straight dissection clamp

97%

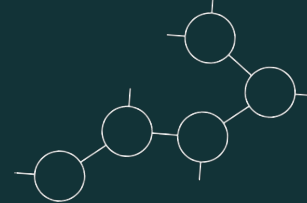
accuracy in
detection of
scalpel

~84%

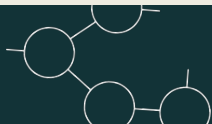
Accuracy in
detection of all
other surgical
tools



Competitive Advantage



		Counting Ability	Time Spent	In-Operative Tracking	Automated System
Scrub XR		✓	< 1 minute	✓	✓
Manual Note Taking		✓	~10 minutes	✓	
SurgiCount+		✓	~5 minutes		
CensiTrac		✓	~5 minutes		



Financials



Manufacturing Cost:
= \$450 per unit



Return on Investment
= 35× return



Retail Price:
= \$1800 per unit



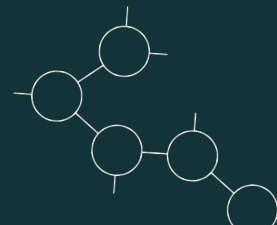
Subscription Model:
= \$12,000 ORs/yr



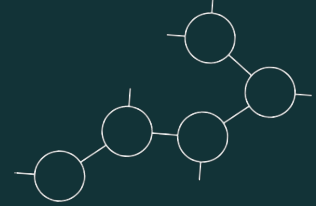
SOM:
\$138K

SAM:
\$1.44M

TAM:
\$703.8M



Values Proposition



1. Efficient: Free up **285 years** of operating room time annually in the U.S.

2. Effective: **Reduce** operating room costs by **~7%** annually at NYP

3. Patient Safety: **Prevent ~2,000** RSI incidents each year, improving patient outcomes



Future Directions

Preference Cards

- Train model to generate surgeon specific preference cards

Dental Procedures

- Applicable even if no risk of RSI
- Increase efficiency of procedures

Conversational AI

- Model trained to audio commands
- Model responds with count of specified tool

Billing Documentation

- Generates dataset that can be used to streamline billing of surgical tools used

Our Team



Sally Scofield

- B.S. Biomedical Engineering and Business Minor
- Cornell University
- Class of 2027
- Lead: DEBUT Project Team



Molly Matri

- B.S. Biomedical Engineering
- Cornell University
- Class of 2027
- Project Manager of DEBUT Project Team



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- B.S. Computer Science and Pre-Medicine
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Rahul Ramarao

- B.S. Computer Science
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- Research & Development Analyst of DEBUT



Pradhi Pakkerakari

- B.S. Computer Science
- Cornell University
- Class of 2027
- Armada, VP ENT, DigAg Project Team (MLE Lead)

Thank you!

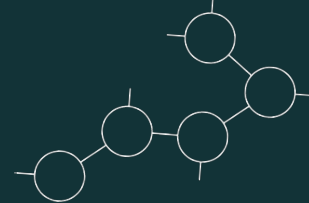


Scrub-XR: Every count counts

Appendix 1: Supporting Literature

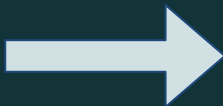
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Appendix 2: Model Training

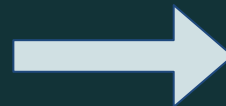


Training Set (1600 Examples):

- Scalpel
- Straight Mayo Scissors
- Curved Mayo Scissors
- Straight Dissection Clamp



**Model
(YOLO v11)**



Evaluation

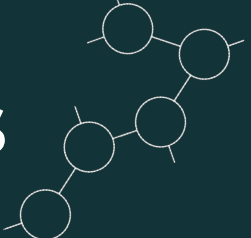


**Validation Set
(200 Examples)**

**Test Set
(200 Examples)**



Appendix 3: Financial Calculations



Manufacturing Cost:

- Estimated from XR headset bill-of-materials benchmarks (camera, processor, optics, sensors, assembly)

Retail Price:

- Set at 4× manufacturing cost, consistent with margins for specialized medical hardware.

Return on Investment:

- 10 min saved per surgery × 3 surgeries/day × \$3,720 OR cost/hr × 262 days ≈ \$487K annual savings per OR
- Total cost of product = \$13,800
- $487,320 / 13,800 = \sim 35$

Software Subscription:

- Benchmarked against surgical workflow and OR software typically priced \$5k–\$20k per operating room annually.

SOM (Serviceable Obtainable Market — Initial Pilot)

- Initial deployment = 10 ORs
- Hardware revenue:
 - 10 ORs × \$1,800 per headset = \$18,000
- Annual subscription revenue:
 - 10 ORs × \$12,000 per OR per year = \$120,000
- Total SOM (first year):
 - \$18,000 + \$120,000 = **\$138,000**

SAM (Serviceable Available Market — NYP System)

- Total NYP operating rooms = 104 ORs
- Hardware revenue:
 - 104 ORs × \$1,800 per headset = \$187,200
- Annual subscription revenue:
 - 104 ORs × \$12,000 per OR per year = \$1,248,000
- Total SAM (first year):
 - \$187,200 + \$1,248,000 = \$1,435,200

TAM (Total Addressable Market — All U.S. Operating Rooms)

- Total U.S. operating rooms ≈ 51,000 ORs
 - Hardware revenue:
 - 51,000 ORs × \$1,800 per headset = \$91,800,000
 - Annual subscription revenue:
 - 51,000 ORs × \$12,000 per OR per year = \$612,000,000
 - Total TAM (first year):
 - \$91,800,000 + \$612,000,000 = \$703,800,000
- 

Appendix 4: Values Prop Calculations

1. Free up 285 years of operating room time annually in the U.S.
 - a. 15 million hospital stays in the U.S. involve operating room (OR) procedures annually
 - b. 10 mins saved per procedure
 - c. $15 \text{ mil} \times 10 \text{ mins} = 285 \text{ years}$

2. Reduce operating room costs by ~7% annually at NYP
 - a. Save 5 mins before and 5 mins after → 10 mins total per surgery
 - b. 3 surgeries a day x 10mins = 30 min OR times saved
 - c. 104 OR for all NYP → \$3,720 per hour x .5 hrs = 193440 per day
 - d. 262 business days → 50.7 million dollars saved annually
 - e. total operating room costs coming in at \$746 million. → reduce by approximately 7% annually